

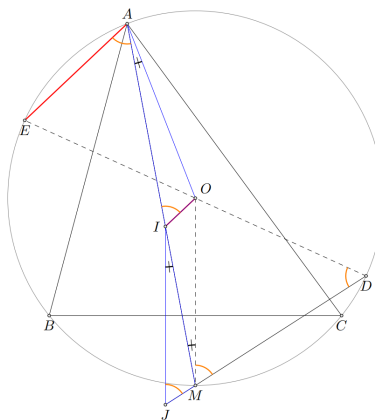
澳門學界數學比賽解答

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摘要

Here is the abstract. As the name suggests, the thing you look at right now is a document which talking about "澳門學界數學比賽解答" 2025. Here I won't post the *Fill in the blank* question. I have nothing more to say to you. Let's watch my solution first!(are you reading this "garbage" abstract? This waste you 10 seconds!)



1 Problem

11. A circle inscribed in a square with side length 2. There is an octagon such that two vertices on each side of square and every side of the octagon is tangent to the circle. Find the possible minimum of the perimeter of the octagon.

12. (JBMO 2022/1) Find all pairs of positive integer (a, b) such that

$$11ab \leq a^3 - b^3 \leq 12ab$$

13. D, E lie on the side AB of an equilateral $\triangle ABC$ such that $AD = DE = EB$. F lie on the side BC such that $CF = AD$. Find the value of $\angle CDF + \angle CEF$.

14. Let a, b, c be non-zero reals. Suppose that $\frac{a^3-1}{a} = \frac{b^3-1}{b} = \frac{c^3-1}{c}$. Find the possible value of $a^3 + b^3 + c^3$.

15. Let a, b, c, d be distinct positive integer such that $\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+d} + \frac{d}{d+a}$ is an integer. Prove that $a + b + c + d$ is not prime.

16. (i) Find the maximum of $x - x^4$, here x is arbitrary real satisfied $0 \leq x \leq 1$.

(ii) Let $a, b, c > 0$ such that $a + b + c = 1$. Prove that

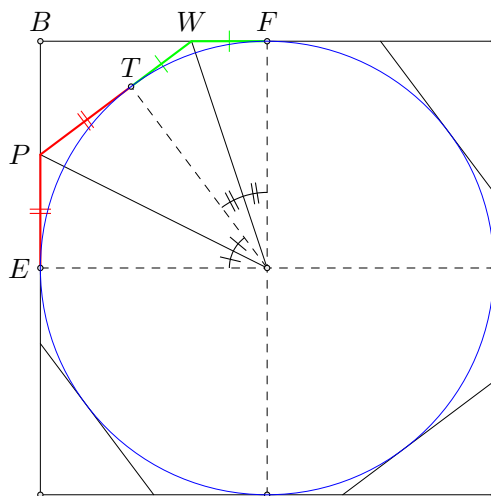
$$a + \sqrt{b} + \sqrt[4]{c} \leq \frac{10+3\sqrt[3]{2}}{8}.$$

2 Solution

11. A circle inscribed in a square with side length 2. There is an octagon such that two vertices on each side of square and every side of the octagon is tangent to the circle. Find the possible minimum of the perimeter of the octagon.

Solution.

We can divide the square into 4 small squares, like the picture below.



Hence, the perimeter of the octagon is $\sum_{cuc} FW + WP + PE$.

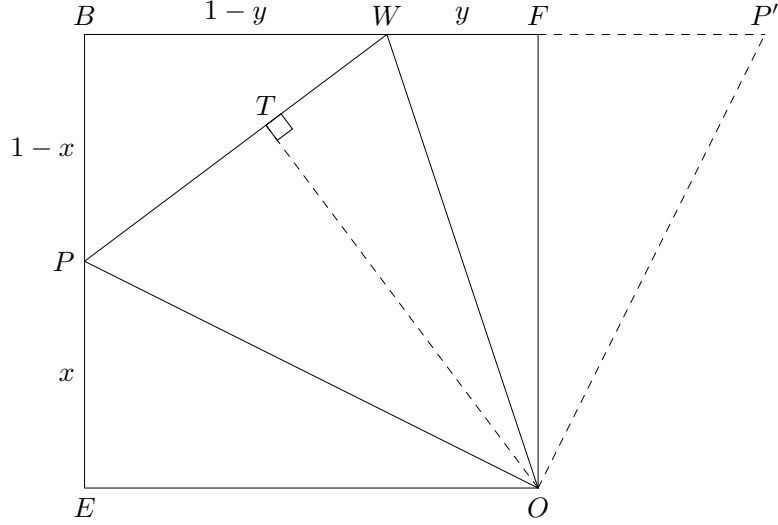
Claim: $WP = EP + FW \iff WP$ is tangent to the circle.

Proof:

Necessity is trivial by using the *Tangent Length Theorem*.

Sufficiency is a classical geometric result. Let's prove it.

As the diagram below shows, extend BF to P' such that $FP' = EP$, then we have $\triangle EOP \cong \triangle FOP'$ (Actually I view this operation as spiral similarity), so $PO = P'O$. Combine with $PW = WF + PE = WF + FP' = WP'$ and $WO = WO$ gives $\triangle WOP \cong \triangle WOP'$, thus $\angle PWO = \angle FWO$.



Now, let T denote the perpendicular feet from O to WP . We can easily get $\triangle TOW \cong \triangle FOW$, which means $OT = OF = \text{radius}$ or T is the tangency point. $\square$

Letting $PE = x, WF = y$, Now there are two approaches to find the minimum of $FW + WP + PE = 2(x + y)$.

First: Deal with the length connection (Propose by Kyle Choi)

Immediately afterwards we have $BP = 1 - x, BW = 1 - y$. Then

$$\begin{aligned} BP^2 + BW^2 &= PW^2 \\ (1 - x)^2 + (1 - y)^2 &= (x + y)^2 \\ (x + 1)(y + 1) &= 2 \end{aligned}$$

Hence $(x + y + 2)^2 \geq 4(x + 1)(y + 1) = 8$ (AM-GM), reduce to $x + y \geq 2\sqrt{2} - 2$. Construct the minimum, that is $x = y = \sqrt{2} - 1$.

Second: Trig Approach

Let $\angle EOP = \alpha, \angle FOW = \beta$, and $\alpha + \beta = \angle POT + \angle TOW = \angle POW = \frac{1}{2}\angle EOF = 45^\circ$. A beautiful thing is that $x = \tan\alpha, y = \tan\beta$. Well,

$$x + y = \tan\alpha + \tan\beta = \frac{\sin\alpha}{\cos\alpha} + \frac{\sin\beta}{\cos\beta} = \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta} = \frac{\sin(\alpha+\beta)}{\frac{1}{2}(\cos(\alpha+\beta) + \cos(\alpha-\beta))} \geq \frac{\sin(\alpha+\beta)}{\frac{1}{2}(\cos(\alpha+\beta) + 1)} = \frac{\sin 45^\circ}{\frac{1}{2}(\cos 45^\circ + 1)} = 2\sqrt{2} - 2.$$

And the construction is the same as First approach.

By the way, the minimum of $\tan\alpha + \tan\beta$ can also use Calculus to "Cal".

Conclusion: $FW + WP + PE \geq 2(2\sqrt{2} - 2)$. Similarly in other small square we have the same thing. Therefore, the perimeter of the octagon: $\sum_{cyc} FW + WP + PE \geq 4 \cdot 2(2\sqrt{2} - 2) = 16\sqrt{2} - 16$. Equality holds iff $EP = WF$ in every small squares.

12. (JBMO 2022/1) Find all pairs of positive integer (a, b) such that

$$11ab \leq a^3 - b^3 \leq 12ab$$

Solution:

$a > b$ holds whatever since $a^3 - b^3 \geq 11ab > 0$.

Let $k = a - b > 0$, and $a^3 - b^3 = k(k^2 + 3ab)$.

Claim: $k \leq 3$.

Proof:

$k(k^2 + 3ab) > 3abk \geq 12ab$ when $k \geq 4$.

Claim: $k > 2$.

Proof:

$k(k^2 + 3ab) \leq 2(2^2 + 3ab) = 6ab + 8 < 11ab$ when $k \leq 2$. (Note: $6ab + 8 < 11ab \iff 5ab > 8$, and $a > b \geq 1$ gives $5ab \geq 5 \cdot 2 \cdot 1 = 10 > 8$)

Let consider $k = 3$.

$11ab \leq 3 \cdot (3^2 + 3ab)$ gives $ab \leq \frac{27}{2}$ reduce to $ab \leq 13$. Same thing doing at $12ab$ gives $ab \geq 9$. Thus, $9 \leq a(a - 3) \leq 13$.

Try to solve the equation $a^2 - 3a - k = 0$ for $k = 9, 10, \dots, 13$ we know only $k = 10$ provide a positive root $a = 5$. In conclude, $(a, b) = (5, 2)$ is the only pair.

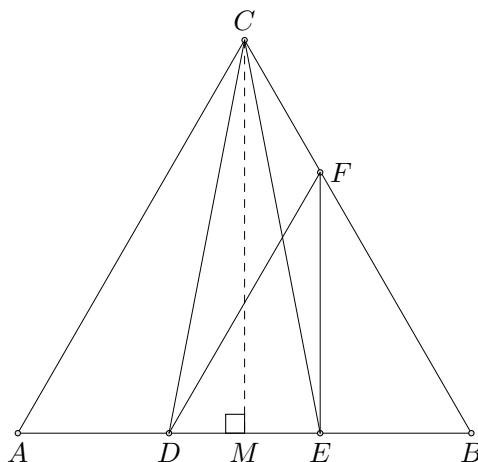
Remark.

$a^3 - b^3 \leq 12ab$ implies that a quadratic is bigger than a cubic, it's very hard to achieve. Trying some (a, b) we realize that $a - b$ should be a small number.

13. D, E lie on the side AB of an equilateral $\triangle ABC$ such that $AD = DE = EB$. F lie on the side BC such that $CF = AD$. Find the value of $\angle CDF + \angle CEF$.

Solution:

Alright, $AB = BC$ and $AD = FC$ gives $CA \parallel FD$, $\angle CDF = \angle ACD$. Furthermore, we have $\triangle BDF$ is equilateral or $FE \perp DB$ since $DE = EB$.



Let M lie on AB such that $CM \perp AB$. Note that CM is the symmetry axis, we have $\angle DCM = \angle ECM$. Meanwhile, $CM \parallel FE$ as they both perpendicular to AB , hence $\angle CEF = \angle ECM$.

The things above implies that $\angle CEF = \angle DCM$. Therefore, $\angle CDF + \angle CEF = \angle ACD + \angle DCM = \angle ACM = 30^\circ$.

14. Let a, b, c be non-zero reals. Suppose that $\frac{a^3-1}{a} = \frac{b^3-1}{b} = \frac{c^3-1}{c}$. Find the possible value of $a^3 + b^3 + c^3$.

Solution:

Existence:

First, we prove that there exist a, b, c such that $\frac{a^3-1}{a} = \frac{b^3-1}{b} = \frac{c^3-1}{c}$ (if not exist, $a^3 + b^3 + c^3$ not exist too).

Just Construct. $(a, b, c) = (2, \frac{2+\sqrt{2}}{2}, \frac{2-\sqrt{2}}{2})$ satisfied $\frac{a^3-1}{a} = \frac{b^3-1}{b} = \frac{c^3-1}{c} = 2$.

Now Letting $\frac{a^3-1}{a} = \frac{b^3-1}{b} = \frac{c^3-1}{c} = k$ where k is real. And we have

$$z^3 - kz - 1 = 0$$

for $z = a, b, c$. Summing these equation gives $a^3 + b^3 + c^3 - (a + b + c)k - 3 = 0$ or $a^3 + b^3 + c^3 = (a + b + c)k + 3$. Since a, b, c are the roots of $z^3 - kz - 1 = 0$, $a + b + c = 0$ must be true by Vieta's Theorem. Thus

$$\boxed{a^3 + b^3 + c^3 = 0 \times k + 3 = 3.}$$

15. Let a, b, c, d be distinct positive integer such that $\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+d} + \frac{d}{d+a}$ is an integer. Prove that $a + b + c + d$ is not prime.

Proof:

Let $T = \frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+d} + \frac{d}{d+a}$.

Determine the range of T anyway, specifically

$$T < \frac{a+b}{a+b} + \frac{b+c}{b+c} + \frac{c+d}{c+d} + \frac{d+a}{d+a} = 4$$

and

$$T > \frac{a}{a+b+c+d} + \frac{b}{b+c+d+a} + \frac{c}{c+d+a+b} + \frac{d}{d+a+b+c} = 1$$

$$T = 4 - \left(\frac{b}{a+b} + \frac{c}{b+c} + \frac{d}{c+d} + \frac{a}{d+a}\right) < 4 - 1 = 3 \text{ (well, same as above).}$$

finally $T = 2$ (Actually "3" is symmetry "1") or $T = \frac{b}{a+b} + \frac{c}{b+c} + \frac{d}{c+d} + \frac{a}{d+a} =$

2. Making a subtraction (A painful expand) gives

$$\begin{aligned} \frac{a-b}{a+b} + \frac{b-c}{b+c} + \frac{c-d}{c+d} + \frac{d-a}{d+a} &= 0 \\ \left(\frac{a-b}{a+b} + \frac{c-d}{c+d}\right) + \left(\frac{b-c}{b+c} + \frac{d-a}{d+a}\right) &= 0 \\ \frac{(a-b)(c+d) + (a+b)(c-d)}{(a+b)(c+d)} + \frac{(b-c)(d+a) + (b+c)(d-a)}{(b+c)(d+a)} &= 0 \\ \frac{2(ac-bd)}{(a+b)(c+d)} + \frac{2(bd-ac)}{(b+c)(d+a)} &= 0 \\ 2(ac-bd)\left(\frac{1}{(a+b)(c+d)} - \frac{1}{(b+c)(d+a)}\right) &= 0 \\ 2(ac-bd)((b+c)(d+a) - (a+b)(c+d)) &= 0 \\ 2(ac-bd)(a-c)(b-d) &= 0 \end{aligned}$$

Since $a \neq b \neq c \neq d$, $ac = bd$ or $a = \frac{bd}{c}$, which means $S = a + b + c + d = \frac{bd}{c} + b + c + d = \frac{bd+bc+c^2+cd}{c} = \frac{(b+c)(c+d)}{c}$ or $S|(b+c)(c+d)$ (We Focus on divisibility).

Now, if S is prime, then $S|b+c$ or $S|c+d$ which gives $S \leq b+c$ or $S \leq c+d$. But $S = a + b + c + d$ is clearly bigger than $b+c$ and $c+d$, contradiction.

16. (i) Find the maximum of $x - x^4$, here x is arbitrary real satisfied $0 \leq x \leq 1$.

(ii) Let $a, b, c > 0$ such that $a + b + c = 1$. Prove that

$$a + \sqrt{b} + \sqrt[4]{c} \leq \frac{10+3\sqrt[3]{2}}{8}.$$

Solution:

(i) the answer is $\frac{3\sqrt[3]{2}}{8}$ by directly differentiate the function $f(x) = x - x^4$. $f(x)$ achieve to $\frac{3\sqrt[3]{2}}{8}$ when $x = \frac{\sqrt[3]{2}}{2}$.

(ii) the function $g(x) = x - x^2$ clearly have the maximum $\frac{1}{4}$, it holds when $x = \frac{1}{2}$. So,

$$a + \sqrt{b} + \sqrt[4]{c} = 1 - b - c + \sqrt{b} + \sqrt[4]{c} = 1 + f(\sqrt[4]{c}) + g(\sqrt{b}) \leq 1 + \frac{1}{4} + \frac{3\sqrt[3]{2}}{8} = \frac{10+3\sqrt[3]{2}}{8}.$$

Equal occurs when $b = \frac{1}{2}$ and $c = \frac{\sqrt[3]{2}}{2}$, meanwhile $a = 1 - b - c$ is positive, we're done.